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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

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2414767823

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **October/November 2010**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

For Examiner's Use

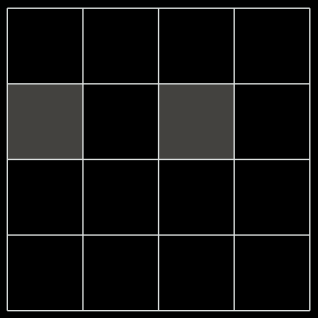
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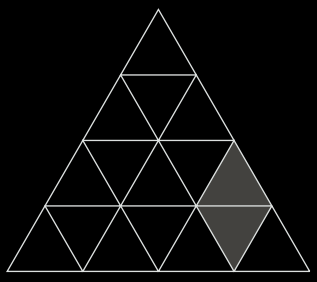


[1]

(b) In the diagram in the answer space, two small triangles are shaded.

Shade four more small triangles, so that the figure will then have rotational symmetry of order 3.

Answer (*b*)



[1]

---

6 The length of a side of an equilateral triangle is given as 41 mm, correct to the nearest millimetre.

(a) Write down the lower bound for the length of a side.

Answer (*a*) ..... mm [1]

(b) Giving your answer in **centimetres**, calculate the lower bound for the perimeter of the triangle.

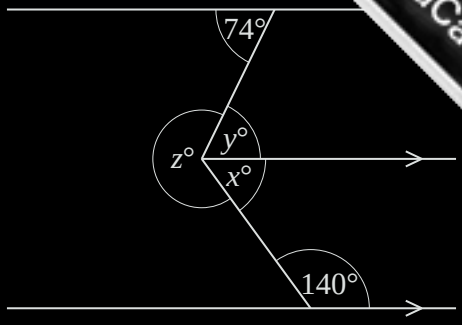
Answer (*b*) ..... cm [1]

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(a) Find x.

Answer (*a*) x = ...[1]

(b) Find y.

Answer (*b*) y = ...[1]

(c) Find z.

Answer (*c*) z = ...[1]

---

12 **(a)** Remove the brackets and simplify $4(7x - 3) - 3(5x - 4)$.

Answer (*a*) .....[1]

(b) Express as a single fraction in its simplest form $\frac{4}{3y} - \frac{5}{4y}$.

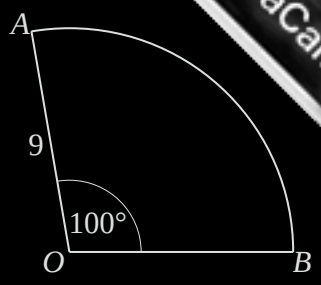
Answer (*b*) .....[1]

(c) Simplify $(4a^2b) \times (3ab^3)$.

Answer (*c*) .....[1]

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Answer (*a*) ..... cm [2]

(b) the perimeter of the sector.

Answer (*b*) ..... cm [1]

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16 The lines $3y + x = 3$ and $y = x + 3$ are shown in the diagram below.

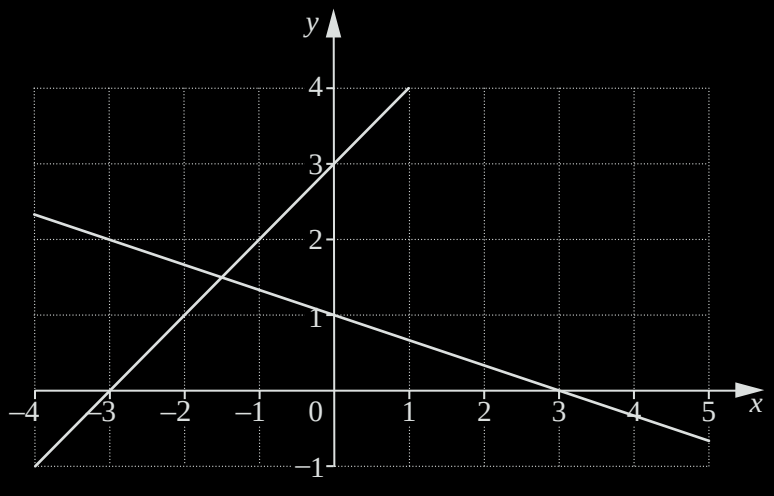
(a) Find the gradient of the line $3y + x = 3$.

Answer (*a*) .....[1]

(b) On the diagram shade, and label with the letter *R*, the region defined by the inequalities

$$3y + x \geqslant 3, \quad y \leqslant x + 3, \quad x \geqslant 0 .$$

Answer (*b*)



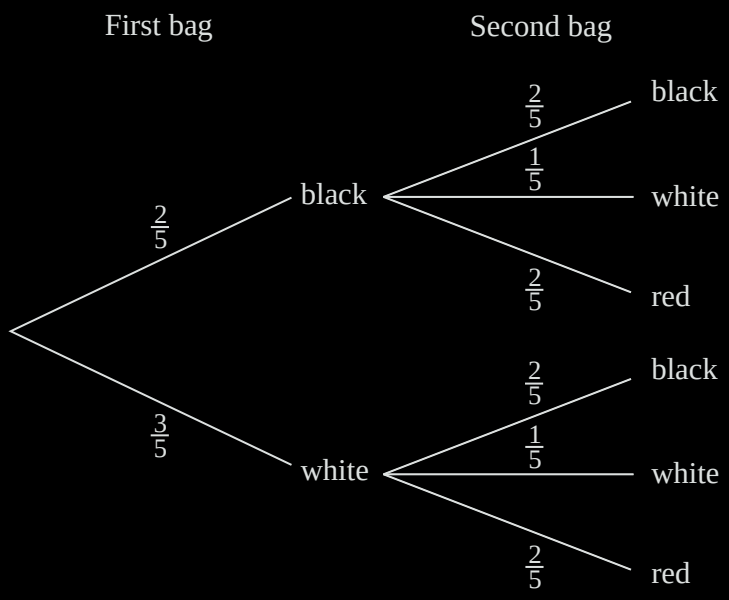
[2]

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Giving each answer in its simplest form, find the probability that

(a) both beads are black,

Answer (*a*) .....[1]

(b) both beads are red,

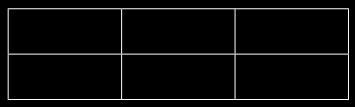
Answer (*b*) .....[1]

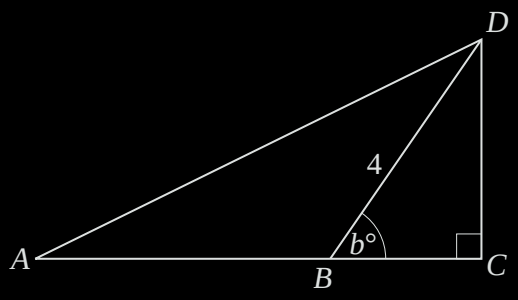
(c) **exactly** one bead is black.

Answer (*c*) .....[2]

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In the diagram, ABC is a straight line.
$BD = 4$ cm, $B\hat{C}D = 90°$ and $C\hat{B}D = b°$.

Use as much information given in the table as is necessary to answer the following questions.

(a) Calculate the value of $\sin A\hat{B}D + \cos A\hat{B}D$.

Answer (*a*) .....[2]

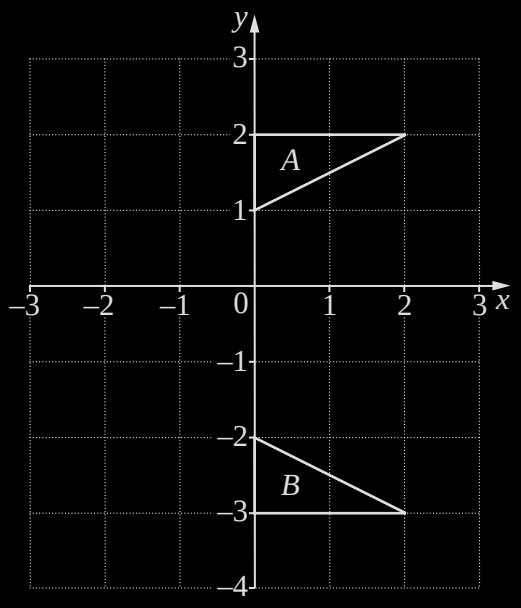
(b) Calculate BC.

Answer (*b*) $BC =$ ..... cm [2]

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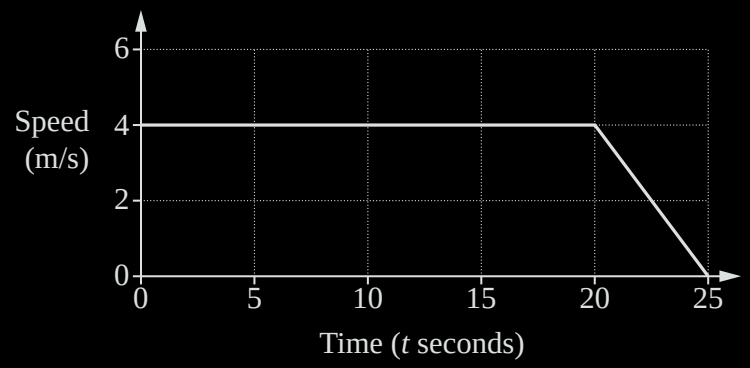


[1]

(ii) Write down the matrix that represents this transformation.

Answer (*b*)(ii) [1]

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The diagram is the speed-time graph of the last 25 seconds of a car's journey.
From $t = 0$ to $t = 20$ the car moves with a constant speed of 4 m/s.
From $t = 20$ to $t = 25$ the car moves with a constant retardation.

(a) Calculate the retardation when $t = 22.5$.

Answer (*a*) ..... m/s^2 [1]

(b) Show that the distance travelled during the 25 seconds is 90 m.

Answer (*b*) .....

.....

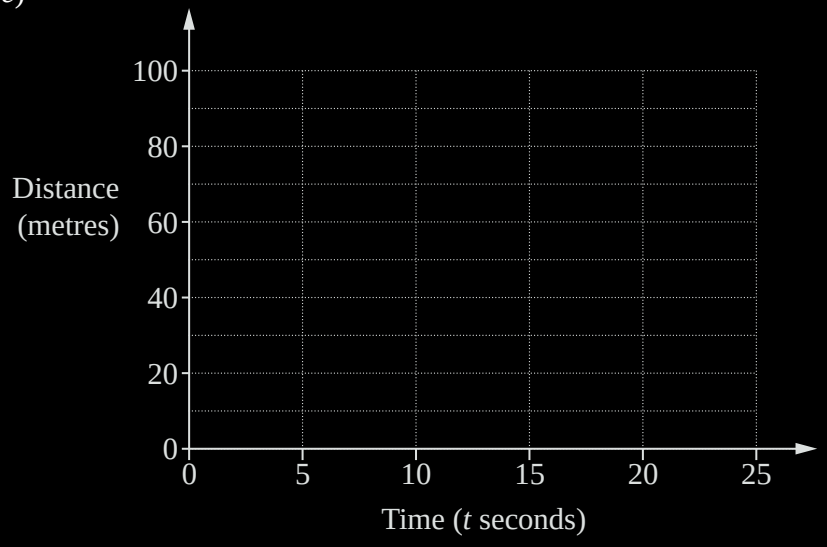
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.....[1]

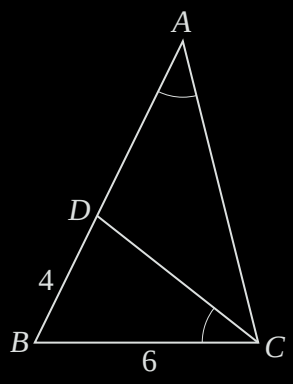
(c) On the grid below, draw the distance-time graph for the 25 seconds.

Answer (*c*)



[2]

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The diagram shows triangle ABC.
D is the point on AB such that $B\hat{C}D = B\hat{A}C$.

(a) Explain why triangle ABC is similar to triangle CBD.

Answer (*a*) .....

.....

.....

.....

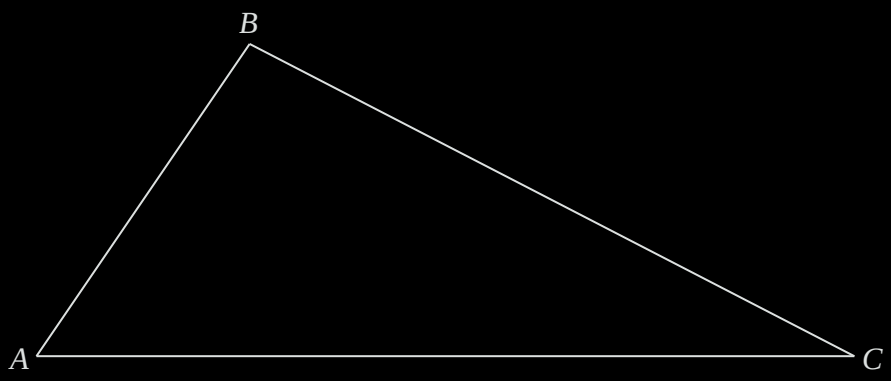
..... [1]

(b) Given that $BD = 4$ cm and $BC = 6$ cm, calculate AD.

Answer (*b*) AD = ... cm [3]

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